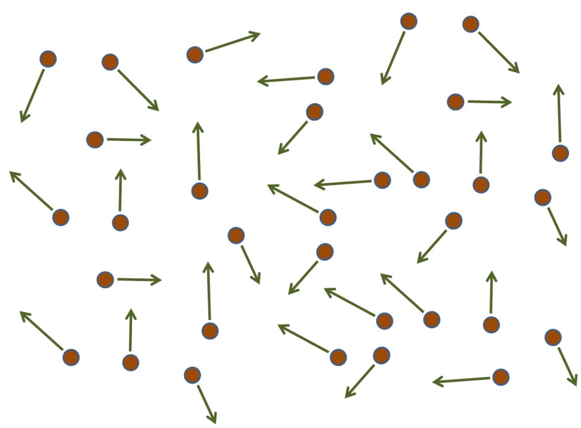
1 Number Distributions

We want to know how many particles we have with a specific size. The AERONET data sets give us a kind of histogram of how may particles we have of a specific size. But the specifics of the graph are more complicated because the data we get is as a function of size so we get something like how many particles there are with a diameter from D to $D + dD$. And it turns out we have done something like this before! Back in PH123 we "studied" the Maxwell Boltzmann distribution of molecular speeds in a gas. It was like a histogram of how many gas molecules had a specific speed. This is similar to what we want. And we will use similar techniques to analyze the data. Let's review.

The internal energy of a system is mostly due to the motion of the molecules in the system. That motion has kinetic energy associated with it. But not a kinetic energy of the whole system moving together. It is a kinetic energy of the parts of the system, the molecules. To find this kinetic energy of the molecules, we would need to know how fast the molecules are going in an ideal gas at a particular temperate and pressure in a particular volume. When we say "speed of the molecules" we have to be specific, what speed? The average? The speed the most particles have?

The average velocity won't do.



The molecule motions have random directions. So the average velocity is zero. Some velocities are in negative directions and some are in positive directions, so the velocities cancel out. But if we square the velocities they are all positive. Then we could find the average of the squared velocities and that would be better. But a velocity squared is not a velocity. So to get back to velocity, let's take a square root.

$$v_{rms} = \sqrt{\overline{v^2}}$$

This quantity is like an average, and gives a representative value that is near the most probable speeds. We have seen this quantity before, it is the rms speed.

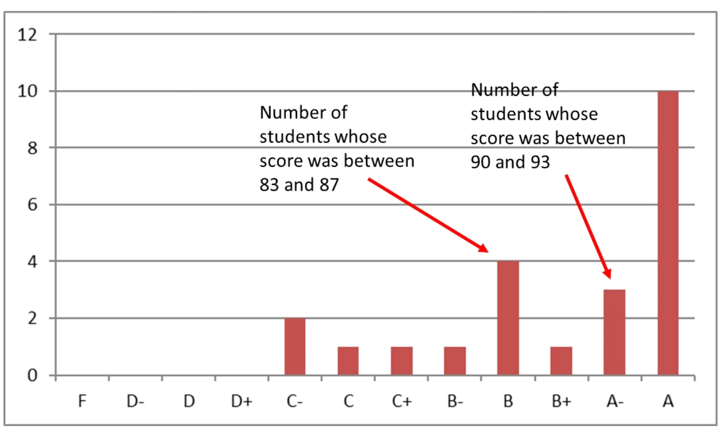
Using this, we can get an expression that relates the temperature of our gas to the speed of the molecules. But if we need to know something about how many molecules have what speed. To find the *distribution* of speeds, let's remember the idea of number density

$$n_V = \frac{\text{\# of molecules}}{\not{V}}$$

We can express the number of molecules this way assuming we know the volume. But we want the speed of the molecules, and we have learned that the concept of energy usually can get us a speed if we don't care to know the direction. Knowing the direction for every air molecule in a room would not be helpful in most cases, so we can be content with just knowing the speed. So lets find a sort of "energy density," that is, the density of molecules that have energy between two amounts of energy, say, E_1 and E_2. We want E_1 and E_2 to be quite close together. So let's let $E_1 = E$ and $E_2 = E + \Delta E$ where ΔE is a small amount of energy. If you take quantum mechanics, or statistical mechanics, you will likely see this quantity quite a lot. Then the number of molecules with a particular energy between E_1 and E_2 could be written as

$$n_V(E)\,dE = \frac{\text{\# of molecules with energy between } E \text{ and } E + \Delta E}{\not{V}} \tag{1}$$

This is called a *distribution function.* We find distribution functions in statistics. They are associated with probabilities. The standard "bell curve" used sometimes in grading is a distribution function. It tells the total number of students that got a particular number of points in a class. Think of grades on a test. If we plot the grades, the height of the bar for a "B" grade is the number of students that scored between 83% and 87%.



This graph is a distribution of grades and it tells us for each score segment, say S and $S + \Delta S$ (think 83% and $83\% + 4\% = 87\%$) how many students had scores in that range. We are doing the same for our molecules and their energy. We will make a graph that shows for each energy segment from E to $E + \Delta E$

how many molecules have that energy. This gives us the probability that the molecules will have a particular energy (or speed, since this is kinetic energy). A function that gives the amount of molecules that have a particular amount of energy also is called a *distribution function* just like it was for grades. The distribution function that we will use is written symbolically in this cryptic fashion, $n_V(E)\,dE$. It is the number of molecules with a particular energy (because our ΔE is so small it is a dE) divided by the total number of molecules. That makes it like a percent.

We won't derive an equation for this quantity, instead we will borrow a result from our junior level thermal physics class.

$$n_V(E) = n_E e^{-\frac{E}{k_B T}} \tag{2}$$

where $n_E dE$ is the number of molecules per unit volume having energy between $E = 0$ and $E = dE$. This distribution function is called the *Boltzmann distribution law.* It tells us that the probability of finding the molecules in a particular energy state varies exponentially as the negative of the energy divided by $k_B T$.

2 Distribution of Molecular Speeds

Since there is a distribution of energies, we expect our gas molecules to have a distribution of velocities. That is the same as saying the molecules in the gas do not all go the same speed. The distribution should depend on temperature, T, since we know the internal energy is tied to temperature. The distribution function is as follows:

$$N_v(v) = 4\pi N \left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2k_B T}} \tag{3}$$

where m is the mass of the molecule. The kinetic energy of the molecules is hiding in the exponent, so we could write this as

$$N_v = 4\pi N \left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} v^2 e^{-\frac{K}{k_B T}} \tag{4}$$

Note that this is of the form for our distribution function because for an ideal gas the only internal energy we have is kinetic energy, $E = K$ so we have $E/(k_B T)$ as the exponent. So we used this to turn our energy distribution function into a speed distribution function.

If there are dN molecules with speeds between v and $v + dv$ then

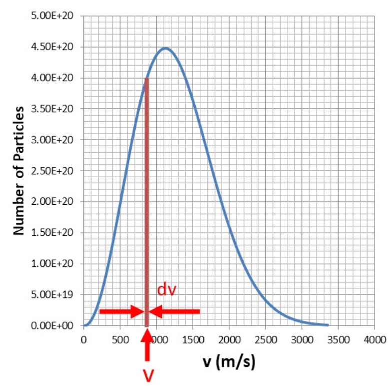
$$dN = N_v dv \tag{5}$$

so there should be

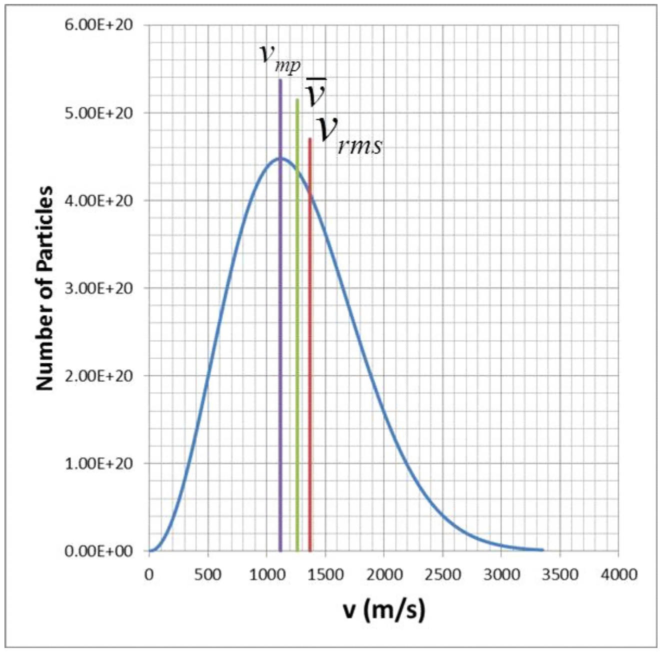
$$N = \int_0^\infty N_v dv \tag{6}$$

total molecules. This tells us that the $4\pi N\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}}$ is a *normalization constant* that makes sure our integral of $N_v dv$ gives us N, the total number of particles.

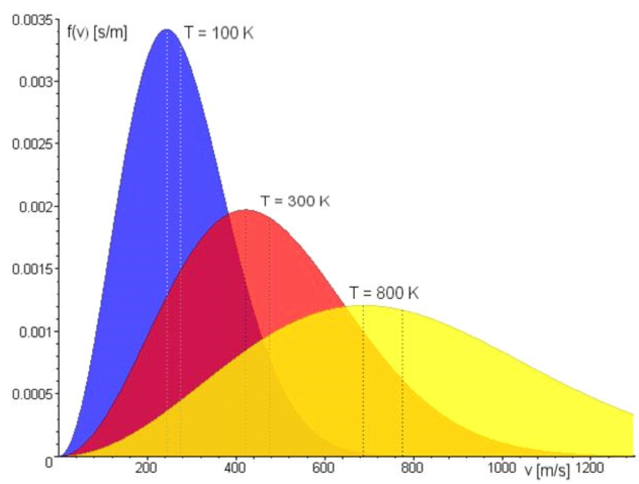
If we plot N_v vs. v we get the figure below. We get a plot that is a little like our grade distribution plot, only for molecular speeds. The number of molecules with speeds between v and $v+dv$ is the area under the blue curve at v. It would be the area of the thin red rectangle right at v with width dv shown in the next figure.



The peak of the curve tells us the most probable speed, that is, the speed the most molecules have, v_{mp} (like in our grade example, the most students got an "A" because that bar was the tallest). The curve is not symmetric, so the most probable speed is not the average speed, $\bar{v}$. There is also our new speed estimate marked v_{rms}.



If we plot N_v for different temperatures, we observe that the peak shifts, and the curve broadens



Temperature dependence of the Maxwell-Boltsman distribution (Image in the Public Domain couracy Fred Stober)

A motivated student could now find the most probably speed by finding the maximum of N_v. We know from our calculus experience how to find a maximum. We take a derivative

$$
\begin{aligned}
\frac{dN_v}{dv} &= \frac{d}{dv}\left(4\pi N\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2k_BT}}\right) \\
&= 4\pi N\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} \frac{d}{dv}\left(v^2 e^{-\frac{m}{2k_BT}v^2}\right) \\
&= 4\pi N\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} \frac{d}{dv}\left(v^2 \frac{d}{dv} e^{-\frac{m}{2k_BT}v^2} + e^{-\frac{m}{2k_BT}v^2}\frac{d}{dv}v^2\right) \\
&= 4\pi N\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} \left(-2\frac{m}{2k_BT}v^3 e^{-\frac{m}{2k_BT}v^2} + 2ve^{-\frac{m}{2k_BT}v^2}\right) \\
&= -2ve^{-\frac{mv^2}{2k_BT}}\left(4\pi N\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}}\right)\left(\frac{mv^2}{2k_BT}-1\right)
\end{aligned}
$$

and set it equal to zero. I didn't spend a lot of time simplifying, because all the factors out front will go away when we set it equal to zero:

$$
\begin{aligned}
0 &= -2ve^{-\frac{mv^2}{2k_BT}}\left(4\pi N\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}}\right)\left(\frac{mv^2}{2k_BT}-1\right) \\
\frac{mv^2}{2k_BT} &= 1 \\
v_{mp} &= \sqrt{\frac{2k_BT}{m}}
\end{aligned}
$$

$$
v_{mp} = \sqrt{\frac{2k_BT}{m}} \tag{7}
$$

and this is great! We have related the temperature, T, to the most probable speed of the molecules. But it is more convenient to use v_{rms}, so let's see if we can modify this expression to be in terms of v_{rms}. And it gives us some insight into using a distribution function.

The average value of v^n is given by

$$
\overline{v^n} = \frac{1}{N}\int_0^\infty v^n N_v dv
$$

Think of this as summing up v^n but using the fact that we already know that for each v there are $N_v dv$ molecules with speed v. So our distribution function makes the integral easier. That motivated student could also use this to find the average speed (not the average velocity, which is zero). He or she will want the average value of v^1

$$
\begin{aligned}
\overline{v^1} &= \frac{1}{N}\int_0^\infty v^1 4\pi N\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2k_BT}} dv \\
&= 4\pi\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}}\int_0^\infty v^3 e^{-\frac{mv^2}{2k_BT}} dv
\end{aligned}
$$

You might be thinking that this integral looks a little hard to do. You would be right. You could use a symbolic math program to do this. Or you can dust off your second semester calculus knowledge. I prefer the symbolic math program, but let's do it the calc II way here. Looking just at the integral

$$\int_0^\infty v^3 e^{-\frac{mv^2}{2k_BT}} dv = \int_0^\infty v^2 e^{-\frac{mv^2}{2k_BT}} v dv$$

Let's do a substitution. let $x = v^2$, $dx = 2vdv$ and $a = -\frac{m}{2k_BT}$ then we have

$$\int_0^\infty v^3 e^{-\frac{mv^2}{2k_BT}} dv = \int_0^\infty x e^{ax} dx$$

and now we have one of those many integration identities that you memorized (and now have forgotten) from Calc II. So we look in an integral table and find

$$\int_0^\infty x e^{ax} dx = \frac{e^{ax}}{a^2}(ax - 1)$$

This matches our integral that we need. We now know the answer. We just have to undo our substitution.

$$\begin{aligned}
\int_0^\infty v^3 e^{-\frac{mv^2}{2k_BT}} dv &= \left.\frac{e^{\frac{-m}{2k_BT}v^2}}{\left(\frac{m}{2k_BT}\right)^2}\left(\left(-\frac{m}{2k_BT}\right)v^2 - 1\right)\right|_0^\infty \\
&= \left[\frac{e^{\frac{-m}{2k_BT}(\infty)^2}}{\left(\frac{m}{2k_BT}\right)^2}\left(\left(-\frac{m}{2k_BT}\right)(\infty)^2 - 1\right)\right] - \left[\frac{e^{\frac{-m}{2k_BT}(0)^2}}{\left(\frac{m}{2k_BT}\right)^2}\left(\left(-\frac{m}{2k_BT}\right)(0)^2 - 1\right)\right] \\
&= 0 - \left[\frac{1}{\left(\frac{m}{2k_BT}\right)^2}\left(\left(-\frac{m}{2k_BT}\right)(0)^2 - 1\right)\right] \\
&= \frac{1}{\left(\frac{m}{2k_BT}\right)^2}
\end{aligned}$$

And we can put our result back in our equation for $\overline{v^1}$

$$
\begin{aligned}
\overline{v^1} &= 4\pi\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}}\int_0^\infty v^3 e^{-\frac{mv^2}{2k_BT}}dv \\
&= 2\pi\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}}\left(\frac{1}{\left(\frac{\pi m}{2\pi k_B T}\right)^2}\right) \\
&= 2\pi\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}}\left(\left(\frac{m}{2\pi k_B T}\right)^{-\frac{4}{2}}\pi^{-2}\right) \\
&= 2\pi\left(\frac{m}{2\pi k_B T}\right)^{=\frac{1}{2}}\left(\pi^{-2}\right) \\
&= 2\left(\frac{2\pi k_B T}{m}\right)^{\frac{1}{2}}\left(\pi^{-1}\right) \\
&= \sqrt{\frac{8k_BT}{\pi m}}
\end{aligned}
$$

so

$$\bar{v} = \sqrt{\frac{8k_BT}{\pi m}}$$

which is also great, but not what we wanted. But it is close. This time let's find the average value of v^2. We hinted that the root mean squared value would be useful. that is, $\overline{v^2} = v_{rms}$. We can find this like we found the average velocity

$$
\begin{aligned}
\overline{v^2} &= \frac{1}{N}\int_0^\infty v^2 4\pi N\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2k_BT}}dv \\
&= 4\pi\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}}\int_0^\infty v^4 e^{-\frac{mv^2}{2k_BT}}dv
\end{aligned}
$$

But the math is a bit more cumbersome. Let

$$a = \frac{m}{2k_BT}$$

Then we have

$$\overline{v^2} = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}}\int_0^\infty v^4 e^{-av^2}dv$$

It is time to look up our integral in a table of integrals. My table tells me that

$$
\begin{aligned}
\int x^4 e^{-ax^2}dx &= -\frac{1}{8a^{\frac{5}{2}}}\left(6\sqrt{a}xe^{-ax^2} - 3\sqrt{\pi}\,\mathrm{erf}\left(\sqrt{a}x\right) + 4a^{\frac{3}{2}}x^3e^{-ax^2}\right)\Big|_0^\infty \\
&= -\frac{1}{8a^{\frac{5}{2}}}\left(-3\sqrt{\pi}\right)
\end{aligned}
$$

The quantity $\operatorname{erf}\left(\sqrt{a}x\right)$ is called the "error function." If you study this function in a good mathematical book on integration you will find that

$$\operatorname{erf}\left(\sqrt{a}\infty\right)=\frac{2}{\sqrt{\pi}}\int_0^{\infty}e^{-t^2}dt=1$$

so

$$\overline{v^2}=4\pi\left(\frac{m}{2\pi k_BT}\right)^{\frac{3}{2}}\left(-\frac{1}{8\left(\frac{m}{2k_BT}\right)^{\frac{5}{2}}}\left(-3\sqrt{\pi}\right)\right)$$

or

$$\overline{v^2}=3\frac{T}{m}k_B$$

now recognize that $v_{rms}=\sqrt{\overline{v^2}}$ so

$$v_{rms}=\sqrt{\frac{3k_BT}{m}} \tag{8}$$

and back in PH123 this is what we wanted. We have an expression that relates the temperature of the gas to the *rms* speed of the molecules of the gas.

We used a speed distribution function to tell us more about our molecular system. We want to do something like this for aerosol particles in the atmosphere. We can use a *size* distribution function to help with our small particles.

2.1 Size Distributions

Suppose we have $\mathbb{N}$ particles *per unit volume* in a sample of air. This is like N but per unit volume

$$\mathbb{N}=\frac{N}{V}$$

and suppose suppose $\mathbb{N}_i$ is the number of particles within a particular diameter range D_i to $D_i+\Delta D_i$. Then we could define a size distribution

$$n_i=\frac{\mathbb{N}_i}{\Delta D_i}$$

This is the number of particles *per unit volume* within a size range divided by that size range so it is the number of particles per meter of size. This would work fine for our particle counter that has specific size bins (like our class grade example that has specific sized grade score bins). But we could make the size of our particle size bins smaller $\Delta D_i\rightarrow dD_i$ so that the bin size is infinitesimal. Then we can define

$$n\left(D\right)dD$$

as the number of particles per unit volume of air having diameters in the range D to $D+dD$. We could take volume in terms of m^3 and we could take diameter in μm. Then the units of $n\left(D\right)$ would be $\frac{1}{\mu\mathrm{m}\,\mathrm{m}^3}$.

For our speed distribution in the last section we found we could integrate our distribution function

$$N = \int_0^\infty N_v dv \tag{9}$$

to get the total number of molecules. This has to work for our size distribution as well

$$\mathbb{N} = \int_0^\infty n\left(D\right) dD \tag{10}$$

Like with our speed distribution, a size distribution is a continuous function. But the number of particles isn't really a continuous quantity. So we have to make sure there are so many particles in our sample that it is reasonable to ignore the discrete nature of particles and treat the collection of particles as though there were some number of them with a specific size D over the entire range from 0 to ∞ although for most of the range the number would be very small, less than one, which doesn't make sense for actual particles. But if we have enough particles, the statistics aren't too different from reality and $n_N\left(D\right) dD$ works to describe our set of particles. For the entire atmosphere, we should be fine. For our particle counter, we will have to stick with $\mathbb{N}_i$ and a particular diameter range D_i to $D_i + \Delta D_i$. But for the whole atmosphere $n_N\left(D\right) dD$ works.

It's customary to define a normalized distribution function.

$$n_N\left(D\right) = \frac{n\left(D\right)}{\mathbb{N}}$$

so that

$$1 = \int_0^\infty n_N\left(D\right) dD \tag{11}$$

that makes $n_N\left(D\right) dD$ the fraction of the total number of particles per m^3 having diameters in the range from D to $D + dD$. And now $n_N\left(D\right)$ is a probability function that gives us the probability of a particle having diameter D.

But let's go back to our un-normalized size distribution. If

$$\mathbb{N} = \int_0^\infty n\left(D\right) dD \tag{12}$$

then

$$d\mathbb{N} = n\left(D\right) dD$$

and we can write

$$n\left(D\right) = \frac{d\mathbb{N}}{dD}$$

and we can even write $n\left(D\right)$ as $d\mathbb{N}/dD$.

2.2 Surface and volume and mass distributions

Having a size distribution function let's us calculate other things, like for our molecular speeds we used the distribution function to find the distribution of kinetic energy. Atmospheric physicists sometimes want the distribution of surface area, volume, or mass for the aerosol particles in the atmosphere. AERONET seems to prefer volume distributions. Let's get expressions for these.[1]

Let's assume our aerosol particles are spherical (they probably are not, but let's assume that anyway). The surface area of a sphere is

$$A_{sphere} = 4\pi r^2$$

and a diameter is

$$D = 2r$$

so

$$A_{sphere} = 4\pi \left(\frac{D}{2}\right)^2 = \pi D^2$$

so we have an area function

$$A(D) = \pi D^2$$

that tells us the are of the particle for every diameter D. And we have our distribution function that tells us how many particles have that diameter in the infinitesimal range D to $D + dD$. So to find the total surface area of all the particles that have diameter D we just multiply the surface area for particles within the range D to $D + dD$ by the number of particles within that range.

$$n_S(D) = \pi D^2 n(D)$$

and if we want the total surface area for all the particles (say you want to know chemical reaction rates for the aerosol so you need the total surface area of all the particles)

$$S = \int_0^\infty n_S(D)\, dD = \int_0^\infty \pi D^2 n(D)\, dD$$

We could do the same thing for the volume. The volume of a sphere is

$$\begin{aligned} V &= \frac{4}{3}\pi r^3 \\ &= \frac{4}{3}\pi \left(\frac{D}{2}\right)^3 \\ &= \frac{4}{3}\pi D^3 \left(\frac{1}{8}\right) \\ &= \frac{\pi}{6} D^3 \end{aligned}$$

and the volume number density would be

$$n_V(D) = \frac{\pi}{6} D^3 n(D)$$

[1] For more on this see Seinfeld and Pandis, *Atmospheric Chemistry and Physics*, 2nd Ed, John Whiley & Sons, Inc. 2006, pp. 358-381

and the volume of all the particles would be

$$V=\int_0^\infty n_V\left(D\right)dD=\int_0^\infty \frac{\pi}{6}D^3 n\left(D\right)dD$$

Mass density makes most sense if all the particles are made of the same material with density ρ

$$n_M\left(D\right)=\rho n_V\left(D\right)=\rho\frac{\pi}{6}D^3 n\left(D\right)$$

and the total mass of all the particles would be

$$m=\int_0^\infty n_m\left(D\right)dD=\int_0^\infty \rho\frac{\pi}{6}D^3 n\left(D\right)dD$$

These size distributions are useful, but hard to plot. This is because the diameters of actual particles may span many orders of magnitude. So a linear plot stretches out a lot and is harder to read. The seemingly obvious solution is to use a semi-log plot so that the D axis is in logarithmic intervals. But if we do this, the area under the curve isn't directly proportional to the number of particles.

The way to fix this is to make a new distribution function where the variable is $\ln\left(D\right)$ or $\log\left(D\right)$ instead of just D. Of course there is a problem with this. D has units! and our logarithm functions have to have dimensionless numbers. You can't take the log of $D=3\,\mu\text{m}$.

$$\ln\left(3\,\mu\text{m}\right)=error$$

To fix this we take $\ln\left(\frac{D}{1\,\mu\text{m}}\right)$ so that if $D=3\,\mu$m. then $\ln\left(\frac{3\,\mu\text{m}}{1\,\mu\text{m}}\right)=\ln\left(3\right)=1.0986$ works. But we have to keep track of the units of our $1\,\mu$m!

Let's define

$$n_e\left(\ln D\right)d\left(\ln D\right)$$

as the number of particles per unit volume of air in the size range $\ln D$ to $\ln D+d\left(\ln D\right)$. It is worth thinking of the units this would have. Because we have divided by $1\,\mu$m inside the logarithms our diameter is now dimensionless. And that makes $\ln D$ dimensionless. So $n_e\left(\ln D\right)$ is just a number density. It has units of one over unit volume. For us that may be $1/\text{m}^3$. In this system the total number of particles would be

$$N=\int_{-\infty}^\infty n_e\left(\ln D\right)d\left(\ln D\right) \tag{13}$$

Note that we had to change the limits of integration because $\ln D$ can be negative. But also note the area under the curve is once again just the concentration of particles! So now looking at the curve tells you easily how many particles of each size there are.

Before we defined size and volume size distributions. We need these again here

$$\begin{aligned} n_{eS}(\ln D) &= \pi D^2 n_e(\ln D) \\ n_{eV}(\ln D) &= \frac{\pi}{6} D^3 n_e(\ln D) \end{aligned}$$

and we need to consider the units for these. The logarithmic size distribution $n_e(\ln D)$ we have already found to have units like $1/\mathrm{m}^3$. And the diameter is likely in units of μm so we pick up $\mu\mathrm{m}^2$

$$n_{eS}(\ln D) = \pi D^2 n_e(\ln D) \qquad \frac{\mu\mathrm{m}^2}{\mathrm{m}^3}$$

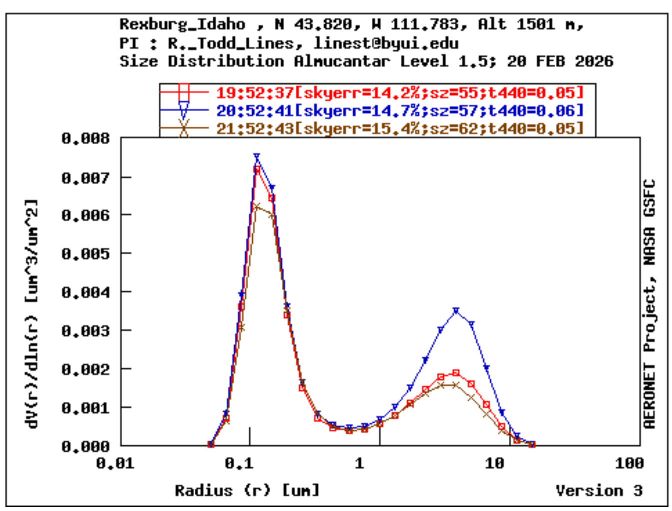
and for the volume distribution

$$n_{eV}(\ln D) = \frac{\pi}{6} D^3 n_e(\ln D) \qquad \frac{\mu\mathrm{m}^3}{\mathrm{m}^3}$$

and

$$\begin{aligned} S &= \int_{-\infty}^{\infty} n_{eS}(\ln D)\, d(\ln D) = \int_0^{\infty} \pi D^2 n_e(\ln D)\, d(\ln D) \\ V &= \int_{-\infty}^{\infty} n_{eV}(\ln D)\, d(\ln D) = \int_0^{\infty} \frac{\pi}{6} D^3 n_e(\ln D)\, d(\ln D) \end{aligned}$$

looking at an AERONET[2] graph we see the vertical axis has units of $\frac{dV(r)}{d(\ln(r))}$ $\left[\frac{\mu\mathrm{m}^3}{\mu\mathrm{m}^2}\right]$



[2] https://aeronet.gsfc.nasa.gov/

Remember that we said

$$n\left(D\right)=\frac{d\mathbb{N}}{dD}$$

From our integral for volume we can realize

$$dV=n_{eV}\left(\ln D\right)d\left(\ln D\right)$$

so then

$$n_{e}\left(\ln D\right)=\frac{dV}{d\left(\ln D\right)}$$

so it appears that the AERONET system is plotting a logarithmic size distribution, but they seem to be plotting as a function of r instead of D. We can fix our equations

Recall that

$$n_{V}\left(D\right)=\frac{\pi}{6}D^{3}n\left(D\right)$$
$$D=2r$$

so

$$\begin{aligned}n_{V}\left(r\right) &= \frac{\pi}{6}\left(2r\right)^{3}n\left(r\right)\\ &= \frac{8\pi}{6}\left(r\right)^{3}n\left(r\right)\\ &= \frac{4\pi}{3}\left(r\right)^{3}n\left(r\right)\end{aligned}$$

so I think we have just

$$n_{eV}\left(\ln r\right)=\frac{4\pi}{3}\left(r\right)^{3}n_{e}\left(\ln r\right)\qquad\frac{\mu\mathrm{m}^{3}}{\mathrm{m}^{3}}$$

but there is one further change for the AERONET. They are collecting all of the particles from the whole atmosphere. This is a total column measurement. So it is as thought we took all the particles from the AERONET path through the whole atmosphere and smashed them down onto a two dimensional surface. We need to integrate over the path through the atmosphere

$$n_{eVc}\left(\ln r\right)=\int_{0}^{TOA}n_{eV}\left(\ln r,z\right)dz\qquad\frac{\mu\mathrm{m}^{3}}{\mathrm{m}^{2}}$$

where TOA means “top of the atmosphere.” I think this is what we see in the AERONET graph. Of course, for the balloon data we need a sum instead of an integral

$$\begin{aligned}n_{eVc}\left(\ln r\right) &= \sum n_{eV}\left(\ln r,z\right)\Delta z\\ &= \sum\frac{4\pi}{3}\left(r\right)^{3}n_{e}\left(\ln r,z\right)\Delta z\end{aligned}$$

because we have discrete numbers. The $n_e(\ln r, z)$ is our size distribution numbers from the balloon flight for a single bin with size r. The Δz would be hour height increments. We could either get these from the altimeter or we could get the assent rate and calculate

$$v_{assent} = \frac{\Delta z}{\Delta t}$$

so

$$\Delta z = v_{assent} * \Delta t$$

and this might change for each altitude so Δz might not be constant.

The sum gives us one condensed (total column) data point for particles with radius r. We need to do this summation for each size bin. So we get a $n_{eVc}(\ln r)$ for each of our sizes. Then the collection of $n_{eVc}(\ln r)$ values should match the shape of the AERONET data. In theory. There are probably some offsets because of the different collection methods and volumes. But hopefully the curve shape is similar.